

3D MODEL · SPEC CONVERSION

E3D Piping Spec Auto Generator from PMC

Analyzes a piping-spec PDF and auto-generates E3D spec MACs. Assembled from catalog · valve family · recipes.

Key Purpose

Why this program

1. Turn a spec PDF into ready E3D spec MACs

- Parses each piping-material class table in the PDF and writes one MAC per spec class

→ **Effect:** Spec build that took days of manual typing is done in seconds with zero transcription error

2. Catch missing catalog names before import

- CATA Check drives an open E3D SPECON console to test every CATREF/DETAIL/BLTREF name, listing any that are undefined

→ **Effect:** Broken spec references are found up front instead of failing silently in the model

3. Compare two spec revisions visually

- Position-aware diff of an old vs new PDF marks new lines RED and changed lines YELLOW on the new PDF

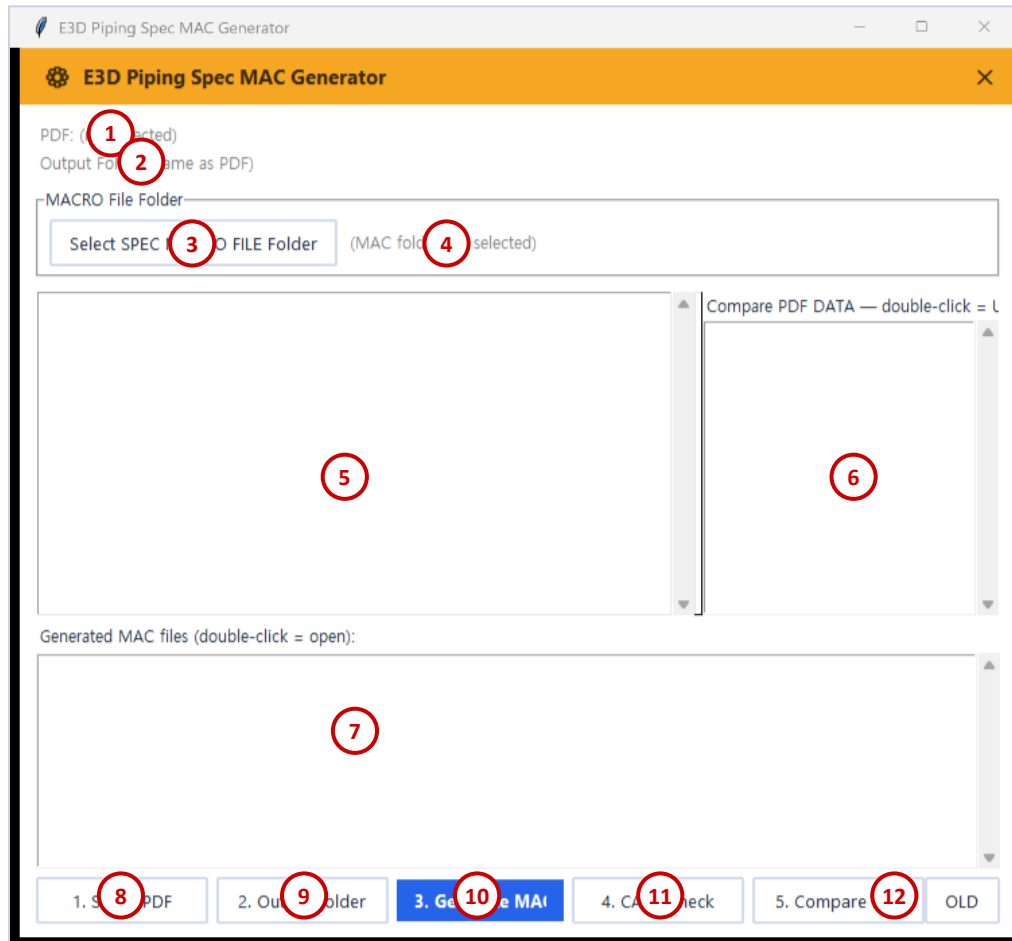
→ **Effect:** Reviewers see exactly what changed between revisions without hand-checking every row

4. Regenerate only the rows that changed

- OLD MAC builds a REV MAC in OLD SPECIFICATION form for just the added/changed sizes per

Required inputs : Piping-spec PDF (for MAC generation); a folder of .MAC files (for CATA Check); two spec PDFs (for Compare / OLD MAC)

Main Window – Item Guide



1. Selected PDF path (not selected yet)
2. Output folder path (defaults to PDF folder)
3. Select SPEC MACRO FILE Folder
4. MAC folder status label
5. Log terminal (progress & results)
6. Compare PDF DATA list (double-click = UltraEdit)
7. Generated MAC files list (double-click = open)
8. 1. Select PDF
9. 2. Output Folder
10. 3. Generate MAC
11. 4. CATA Check
12. 5. Compare PDF / OLD MAC

Overview

What it does

1. Analyzes a piping-spec PDF and auto-generates E3D spec MACs
2. Assembled from catalog
3. Valve family
4. Recipes

Category: 3D Model · Spec Conversion

Folder: E3D_SPEC_GEN

Main scripts: spec_gen.py, pdms_launcher.py

Scripts / Status: 21 scripts · Operational

Tags: E3D · SPEC · MAC · PDF · catalog

Output Samples

Output Samples

PIPING MATERIAL CLASSES							
DESIGN CONDITION	LINE SERVICE	PDC	PMC	HBD1			
ACCORDING TO INDEX	LP STEAM	LCO	CLASS RATING	CL 150			
	CRD TANK DRAIN (AFTER BLOCK V/V TO IBD TANK)	LGA	BASE MATERIAL	CARBON STEEL			
	START-UP BLENDER DRAIN	HGA	CODE	ASME B31.1			
	ABNORMAL BLENDER DRAIN	HAN	CORROSION ALLOWANCE(mm)	1.6			
	ROUILLI BOTTOM HEADER DRAIN(AFTER MOV TO IBD TANK)	LCO					
	IBD TANK OVERFLOW TO PIT	LCO					
	IBD TANK LEVEL GAUGE & TRANSMITTER CONNECTION	NA					
	IBD TANK PUMP SUCTION / DISCHARGE	LCO					
	PSV DISCHARGE & EXHAUST VENT FOR STEAM DRUM	HAD					
	PSV DRIP PAN DRAIN	HAD					
	SUPERHEATER DRAIN (AFTER MOV)	HAD/NAH					
	STEAM ZONE DRAIN HEADER (TO IBD TANK)	LCO					
FUEL GAS SUPPLY TO FCV	HFG						
FUEL GAS SUPPLY FROM FCV TO BURNER, VENTILATION	HFG						
IGNITION GAS	H-G						
Cooling WATER SUPPLY & RETURN	POB						
NO GAS	GJA						
PIPE AND FITTING				VALVE			
ITEM	SIZE(DN)	SCH / RATING	MATERIAL	DESCRIPTION	SIZE(DN)	MATERIAL	Remark
PIPES	15-50	S-80	ASTM A106 GR.B	SMLS, PE	GATE		
	95-250	S-40	ASTM A106 GR.B	SMLS, BE	15-50	CL 600, ASTM A105, SWE	
	300-350	S-STD	ASTM A106 GR.B	SMLS, BE	95-350	CL 150, ASTM A216 GR WCB, BW	
					15-350 (Note1)	CL 150, ASTM A216 GR WCB, FLANGED RF	
				GLOBE			
					15-50	CL 600, ASTM A105, SWE	
FITTINGS	15-50	CL 3000	ASTM A105	SWE	95-350	CL 150, ASTM A216 GR WCB, BW	
	95-250	S-40	ASTM A234 GR WPB	SMLS, BW	15-350 (Note1)	CL 150, ASTM A216 GR WCB, FLANGED RF	
	300-350	S-STD	ASTM A234 GR WPB	SMLS, BW	CHECK		
					15-50	CL 600, ASTM A105, SWE	
					95-350	CL 150, ASTM A216 GR WCB, BW	
				BALL			
FLANGES	15-50	CL 150	ASTM A105	RF, SWE	15-50	CL 600, ASTM A105, SWE	
	95-350	CL 150	ASTM A105	RF, WN, BW	95-350	CL 150, ASTM A216 GR WCB, FLANGED RF	
GASKETS	ALL	CL 150	SPIRAL WOUND 4 MM THK, SS304, PTFE/RTG/TER INCL. FILLER GRANULES		15-350 (Note1)	CL 150, ASTM A216 GR WCB, FLANGED RF	
BOLT / NUT	ALL		STUD BOLT : ASTM A193 GR B7 (H.D Galv.)	BUTTERFLY			
	ALL		HEAVY HEX NUT : ASTM A194 Gr 2H (H.D Galv.)	95-600	CL 150, ASTM A216 GR WCB, LUGGED RF		
NOTE 1. Flanged type valves shall be applied for special case according to detail design. (Control valve, On-off valve, Mechanical package applied valve, Solenoid valve, Level gauge, etc)							

PMS Spec Sample

E3D Spec Auto Gen from PMC sample									
ITEM	SIZE(DN)	SCH / RATING	MATERIAL	DESCRIPTION	SIZE(DN)	MATERIAL	ITEM	SIZE(DN)	SCH / RATING
PIPE	15-50	S-80	ASTM A106 GR.B	SMLS, PE	GATE		PIPE	15-50	S-80
PIPE	95-250	S-40	ASTM A106 GR.B	SMLS, BE	15-50	CL 600, ASTM A105, SWE	PIPE	95-250	S-40
PIPE	300-350	S-STD	ASTM A106 GR.B	SMLS, BE	95-350	CL 150, ASTM A216 GR WCB, BW	PIPE	300-350	S-STD
FITTING					15-350 (Note1)	CL 150, ASTM A216 GR WCB, FLANGED RF	FITTING		
FITTING				GLOBE			FITTING		
FITTING					15-50	CL 600, ASTM A105, SWE	FITTING		
FITTING	15-50	CL 3000	ASTM A105	SWE	95-350	CL 150, ASTM A216 GR WCB, BW	FITTING	15-50	CL 3000
FITTING	95-250	S-40	ASTM A234 GR WPB	SMLS, BW	15-350 (Note1)	CL 150, ASTM A216 GR WCB, FLANGED RF	FITTING	95-250	S-40
FITTING	300-350	S-STD	ASTM A234 GR WPB	SMLS, BW	CHECK		FITTING	300-350	S-STD
FITTING					15-50	CL 600, ASTM A105, SWE	FITTING		
FITTING					95-350	CL 150, ASTM A216 GR WCB, BW	FITTING		
FITTING				BALL			FITTING		
FLANGE	15-50	CL 150	ASTM A105	RF, SWE	15-50	CL 600, ASTM A105, SWE	FLANGE	15-50	CL 150
FLANGE	95-350	CL 150	ASTM A105	RF, WN, BW	95-350	CL 150, ASTM A216 GR WCB, FLANGED RF	FLANGE	95-350	CL 150
GASKET	ALL	CL 150	SPIRAL WOUND 4 MM THK, SS304, PTFE/RTG/TER INCL. FILLER GRANULES		15-350 (Note1)	CL 150, ASTM A216 GR WCB, FLANGED RF	GASKET	ALL	CL 150
BOLT / NUT	ALL		STUD BOLT : ASTM A193 GR L87 (H.D Galv.)	BUTTERFLY			BOLT / NUT	ALL	
BOLT / NUT	ALL		HEAVY HEX NUT : ASTM A194 Gr 2H (H.D Galv.)	95-600	CL 150, ASTM A216 GR WCB, LUGGED RF		BOLT / NUT	ALL	

E3D Spec Auto Gen from PMC sample

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: 3D Model · Spec Conversion.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: spec_gen.py, pdms_launcher.py.
- Output samples and the program window above reflect the current build.

Thank You

E3D Piping Spec Auto Generator from PMC · Plant Engineering Total Service System